
Logic and Computer Design Fundamentals

Chapter 2 – Combinational Logic Circuits

Part 1 – Gate Circuits and Boolean Equations

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Overview

- **Part 1 – Gate Circuits and Boolean Equations**
 - Binary Logic and Gates
 - Boolean Algebra
 - Standard Forms
- **Part 2 – Circuit Optimization**
 - Two-Level Optimization
 - Map Manipulation
 - Multi-Level Circuit Optimization
- **Part 3 – Additional Gates and Circuits**
 - Other Gate Types
 - Exclusive-OR Operator and Gates
- **Part 4 – HDLs overview**
 - Logic Synthesis
 - HDL Representations—Verilog

Binary Logic and Gates

- Binary variables (Boolean variables) take on one of two values (e.g., 0 and 1).
- Logical operators operate on binary values and binary variables.
- Basic logical operators are the logic functions AND, OR and NOT.
- Logic gates implement logic functions.
- Boolean Algebra (Logical Algebra) is a branch of mathematics that deals with Binary variables.
- We study Boolean algebra as a foundation for designing and analyzing digital systems!

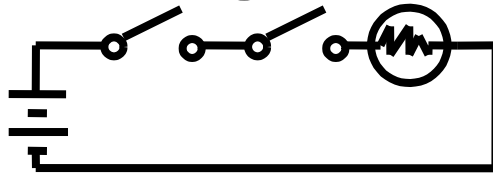
Binary Variables

- **Recall that the two binary values have different names:**
 - **True/False**
 - **On/Off**
 - **Yes/No**
 - **1/0**
- **We use 1 and 0 to denote the two values.**
- **Variable identifier examples:**
 - **A, B, y, z, or X_1 for now**
 - **RESET, START_IT, or ADD1 later**

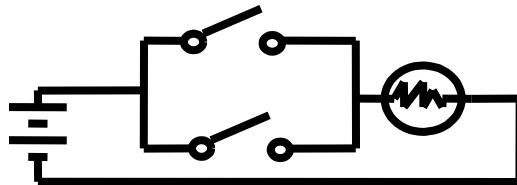
Logical Operations

- The three basic logical operations are:

- AND



- OR



- NOT

- AND is denoted by a dot ($\cdot$).
- OR is denoted by a plus ($+$).
- NOT is denoted by an overbar ($\bar{}$), a single quote mark ($'$) after, or ($\sim$) before the variable.

Notation Examples

- **Examples:**

- $Y = A \times B$ is read “Y is equal to A AND B.”
- $z = x + y$ is read “z is equal to x OR y.”
- $X = \bar{A}$ is read “X is equal to NOT A.”

- **Note: The statement in algebra:**

$1 + 1 = 2$ (read “one plus one equals two”)

is not the same as in **Boolean algebra**

$1 + 1 = 1$ (read “1 or 1 equals 1”).

Operator Definitions

- Operations in Boolean algebra are as defined on the values "0" and "1" for each operator:

AND

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

OR

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 1$$

NOT

$$\overline{0} = 1$$

$$\overline{1} = 0$$

Truth Tables

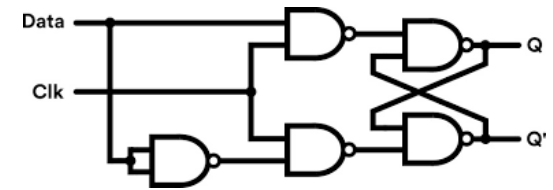
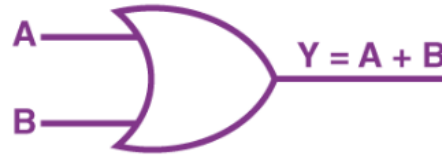
- *Truth table* – a tabular listing of the values of a function for all possible combinations of values on its arguments
- Example: Truth tables for the basic logic operations:

AND		
X	Y	$Z = X \cdot Y$
0	0	0
0	1	0
1	0	0
1	1	1

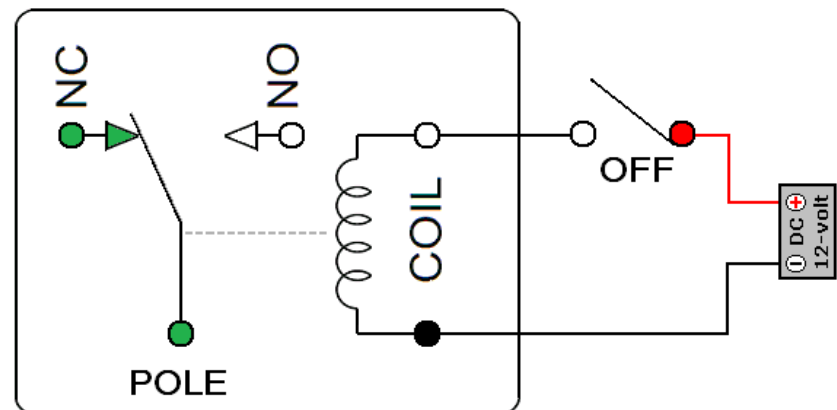
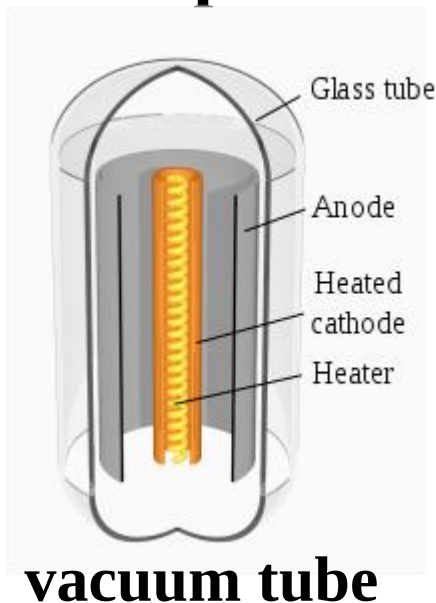
OR		
X	Y	$Z = X + Y$
0	0	0
0	1	1
1	0	1
1	1	1

NOT	
X	$Z = \overline{X}$
0	1
1	0

Logic Gates



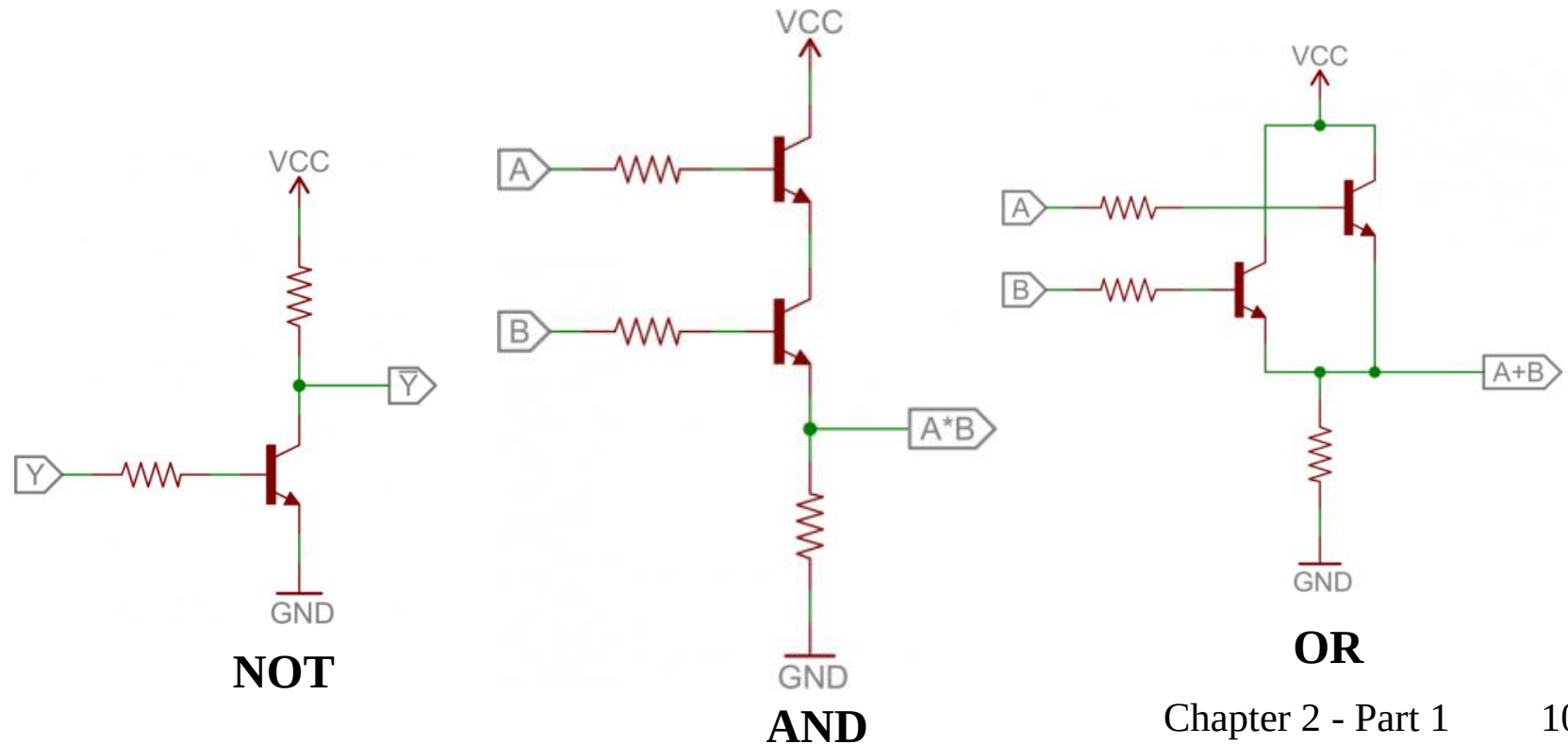
- In the earliest computers, switches were opened and closed by magnetic fields produced by energizing coils in *relays*. The switches in turn opened and closed the current paths.
- Later, *vacuum tubes* that open and close current paths electronically replaced relays.



relay

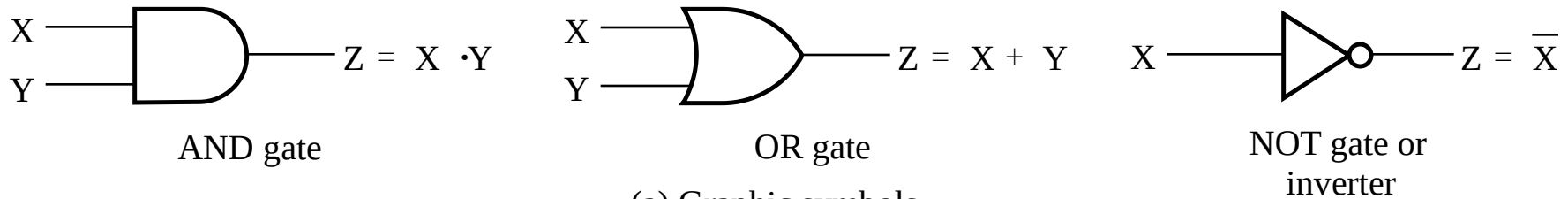
Logic Gates (Continued)

- Today, *transistors* are used as electronic switches that open and close current paths.
- Implementation of logic gates with bi-polar junction transistor (BJT)



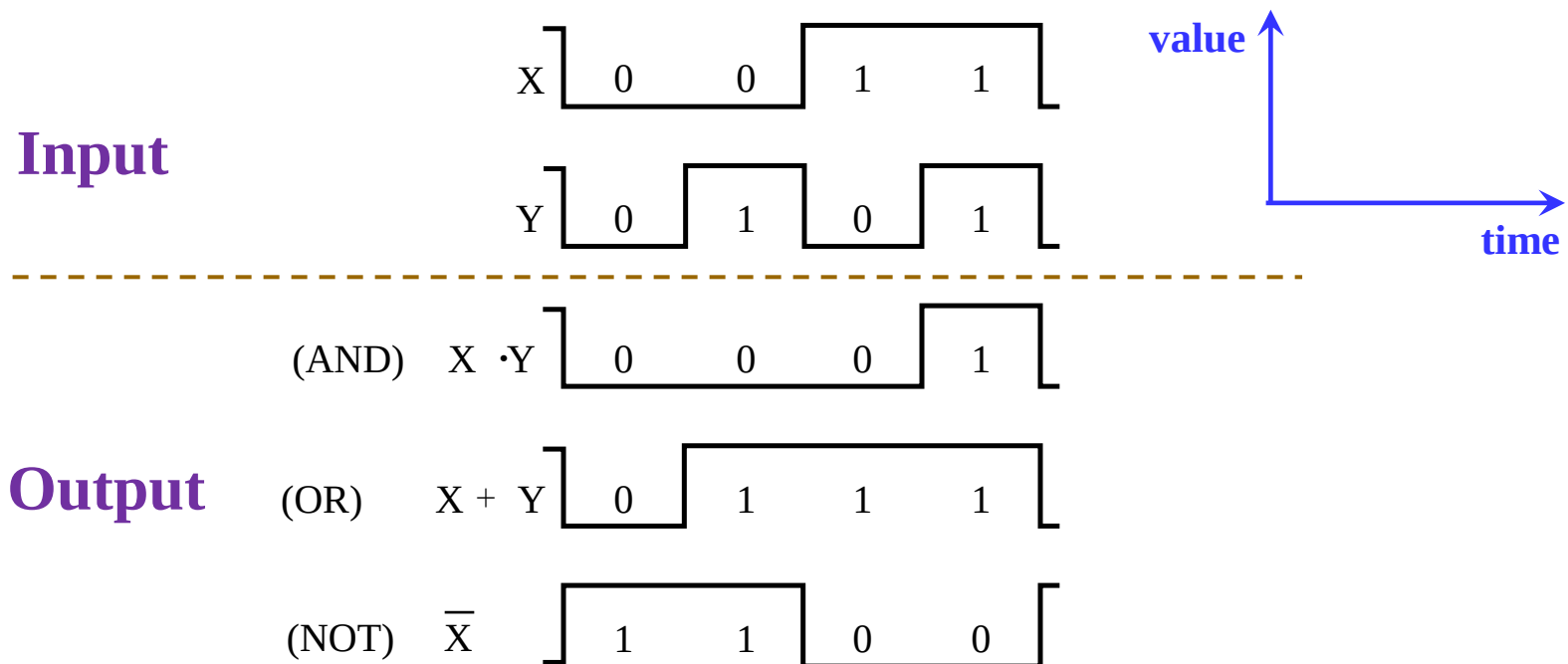
Logic Gate Symbols and Behavior

- Logic gates have special symbols:



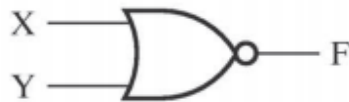
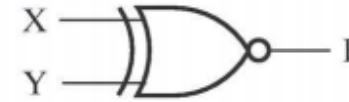
(a) Graphic symbols

- Waveform of logic gates are as follows:



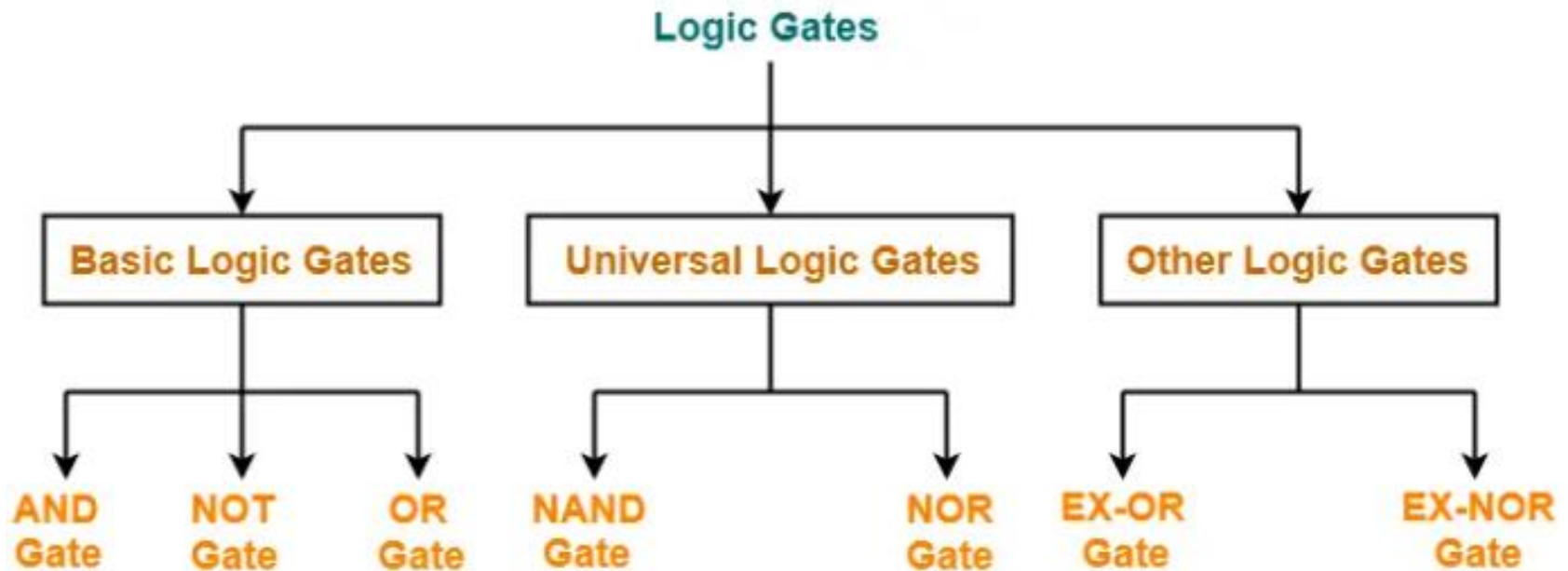
(b) Timing diagram

Other Commonly Used Logic Gates

NAND		$F = \overline{X \cdot Y}$	<table><tr><th>X</th><th>Y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	X	Y	F	0	0	1	0	1	1	1	0	1	1	1	0
X	Y	F																
0	0	1																
0	1	1																
1	0	1																
1	1	0																
NOR		$F = \overline{X + Y}$	<table><tr><th>X</th><th>Y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	X	Y	F	0	0	1	0	1	0	1	0	0	1	1	0
X	Y	F																
0	0	1																
0	1	0																
1	0	0																
1	1	0																
Exclusive-OR (XOR)		$F = X\overline{Y} + \overline{X}Y$ $= X \oplus Y$	<table><tr><th>X</th><th>Y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	X	Y	F	0	0	0	0	1	1	1	0	1	1	1	0
X	Y	F																
0	0	0																
0	1	1																
1	0	1																
1	1	0																
Exclusive-NOR (XNOR)		$F = \overline{XY + \overline{X}\overline{Y}}$ $= X \oplus Y$	<table><tr><th>X</th><th>Y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	X	Y	F	0	0	1	0	1	0	1	0	0	1	1	1
X	Y	F																
0	0	1																
0	1	0																
1	0	0																
1	1	1																

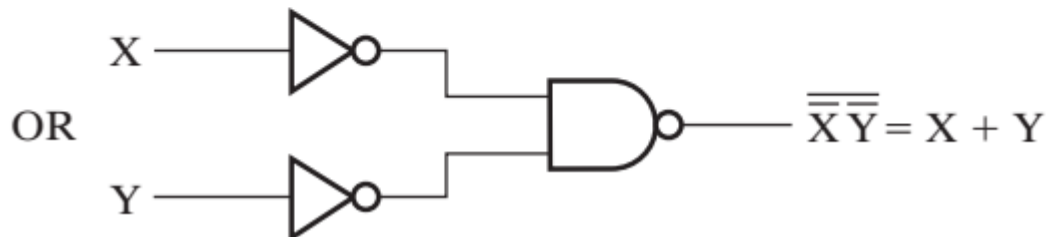
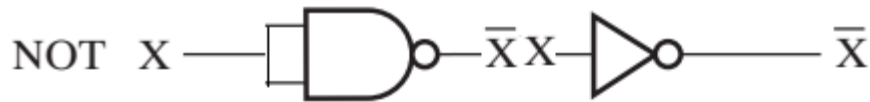
Types of Logic Gates

- Logic gates can be classified into three different types.



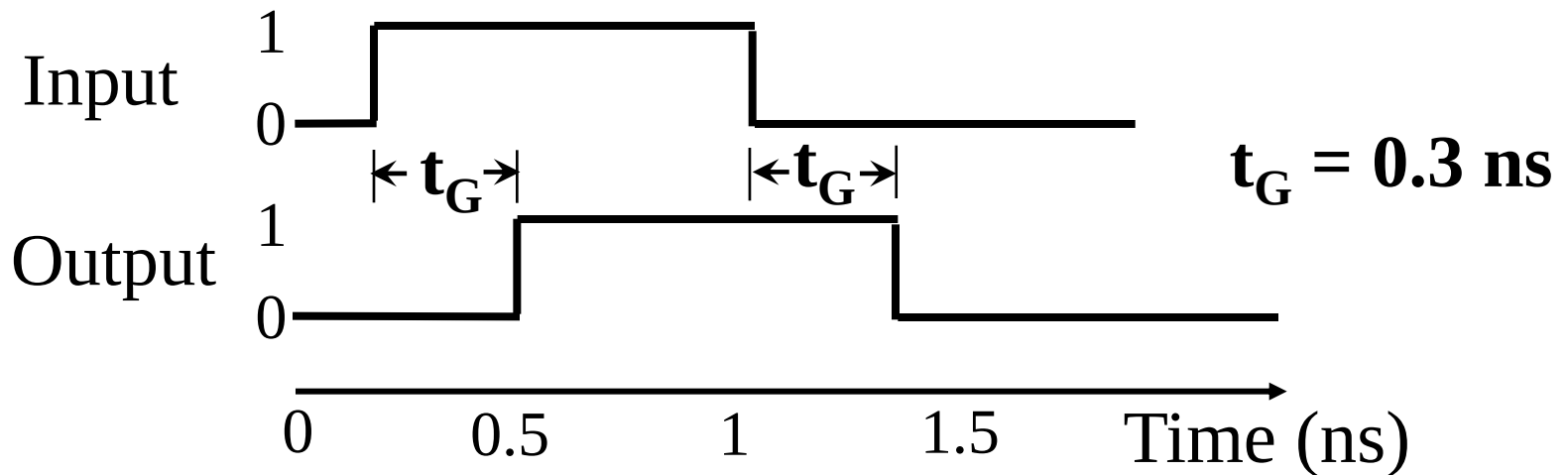
Universal Gate

- A gate type that alone can be used to implement all possible Boolean functions is called a **universal gate** and is said to be “**functionally complete**”.
- For example, NAND gate is a universal gate.



Gate Delay

- In actual physical gates, if one or more input changes cause the output to change, the output change does not occur instantaneously.
- The delay between an input change(s) and the resulting output change is the *gate delay* denoted by t_G :



Logic Diagrams and Expressions

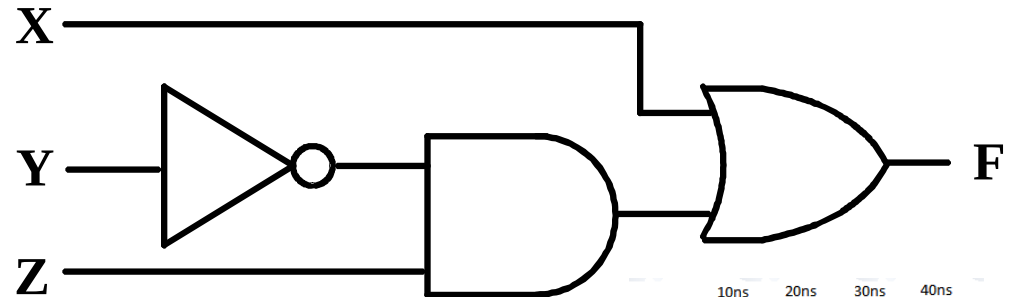
① Truth Table

X Y Z	$F = X + \bar{Y} \times Z$
0 0 0	0
0 0 1	1
0 1 0	0
0 1 1	0
1 0 0	1
1 0 1	1
1 1 0	1
1 1 1	1

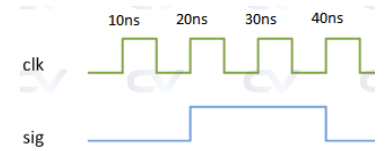
② Equation

$$F = X + \bar{Y} Z$$

③ Logic Diagram



④ Waveform



- Boolean equations, truth tables, logic diagrams and waveform describe the same function!
- **Truth tables and waveforms are unique; equations and diagrams are not.**
- This gives flexibility in implementing functions.

Boolean Algebra

- **An algebraic structure** defined on a set of **at least two elements**, B (for boolean algebra, $B=\{0, 1\}$), together with **three binary operators** (denoted $+$, $\cdot$ and $\overline{}$) that satisfies the following basic **identities**:
-

1. $X + 0 = X$

2. $X \cdot 1 = X$

3. $X + 1 = 1$

4. $X \cdot 0 = 0$

5. $X + X = X$

6. $X \cdot X = X$

7. $X + \overline{X} = 1$

8. $X \cdot \overline{X} = 0$

9. $\overline{\overline{X}} = X$

10. $X + Y = Y + X$

11. $XY = YX$

Commutative

12. $(X + Y) + Z = X + (Y + Z)$

13. $(XY)Z = X(YZ)$

Associative

14. $X(Y + Z) = XY + XZ$

15. $X + YZ = (X + Y)(X + Z)$

Distributive

16. $\overline{X + Y} = \overline{X} \cdot \overline{Y}$

17. $\overline{X \cdot Y} = \overline{X} + \overline{Y}$

DeMorgan's

Some Properties of Identities & the Algebra

- If the meaning is unambiguous, we leave out the symbol “.”
- The identities above are organized into pairs. These pairs have names as follows:

1-4 Existence of 0 and 1

5-6 Idempotence

7-8 Existence of complement

9 Involution

10-11 Commutative Laws

12-13 Associative Laws

14-15 Distributive Laws

16-17 DeMorgan's Laws

Boolean Operator Precedence

- **The order of evaluation in a Boolean expression is:**
 1. Parentheses
 2. NOT
 3. AND
 4. OR
- **Consequence: Parentheses appear around OR expressions**
- **Example:** $F = A(B + C)(C + \overline{D})$

Advanced Boolean Theorems

■ Duality Theorem

- The Duality Theorem states that every Boolean expression remains valid if the operators and identity elements are swapped ($\text{AND} \leftrightarrow \text{OR}$, $0 \leftrightarrow 1$).

■ Substitution Theorem

- Replacing expressions in a Boolean equation with their logically equivalent expressions preserves the logical equivalence of the equation.

■ Complementary Theorem

- The Complementary Theorem deals with the behavior of Boolean expressions involving variables and their complements.

Duality Theorem

- The **dual of an algebraic expression** is obtained by:
 - **interchanging AND (+) and OR (·)**
 - **interchanging 0 and 1**notice: with variables remaining unchanged!
- Seek the dual of a function, **the operation sequence keep as same as the original function.**
- The identities appear in dual pairs. When there is only one identity on a line the identity is self-dual (i.e., the dual expression = the original expression).
- Unless it happens to be self-dual, the dual of an expression does not equal the expression itself.

Duality Theorem (continued)

- **Example:** $F = (A + \bar{C}) \cdot B + 0$
 $\text{dual } F = (A \cdot \bar{C} + B) \cdot 1 = A \cdot \bar{C} + B$
- **Example:** $G = X \cdot Y + \overline{(W + Z)}$
 $\text{dual } G = ((X+Y) \cdot \overline{(W \cdot Z)}) = (X+Y) \cdot (\bar{W} + \bar{Z})$
- **Example:** $H = A \cdot B + A \cdot C + B \cdot C$
 $\text{dual } H = (A + B) \cdot (A + C) \cdot (B + C)$
- **Are any of these functions self-dual?**

Duality Theorem (continued)

- If the function G is the dual of F , then F is also G of duality. G and F is mutually duality formula.
- If the two logical functions F and G are equal ($F=G$), then the duality formula F' and G' are also equal ($F' = G'$).

- Example

$$X + XY = X \quad \longrightarrow \quad X(X+Y) = X \quad \text{Absorption}$$

$$X(Y+Z) = XY + XZ \quad \longrightarrow \quad X + YZ = (X+Y)(X+Z) \quad \text{Distributive}$$

Substitution Theorem

■ Any logical equation that contains a variable **A**, and if all occurrences of **A's position** are replaced with a logical function **F**, the equation still holds.

■ Example: $X(Y+Z)=XY+XZ$, if $X + YZ$ instead of X , then the equation still holds:

$$(X+YZ)(Y+Z)=(X+YZ)Y+(X+YZ)Z$$

Complementary Theorem

- The Complementary Theorem describes the relationship between a variable and its complement.

- $A \cdot \overline{A} = 0$

- $A + \overline{A} = 1$

- The inverse theorem of Complementary Theorem

- If two variables A and B satisfy the conditions of the Complementary Theorem, then B must be the complement of A (i.e., $B = \overline{A}$).

- $A \cdot B = 0$
 - $A + B = 1$

}  $B = \overline{A}$

Complementing Functions

- For logic function F , the **inverse function** of the original function is obtained by:
 - **interchanging AND (+) and OR (·)**
 - **complementing** each **constant value** and **literal**
- The inverse function of the original function is referred to as: $\bar{F}$, F' or $\sim F$.

■ Example

$$F = \bar{A}B + C\bar{D} \quad \longrightarrow \quad \bar{F} = (A + \bar{B})(\bar{C} + D)$$

Complementing Functions (continued)

- Use **DeMorgan's Theorem** to complement a function.
- **DeMorgan's Theorem**

$$\overline{A \cdot B} = \overline{A} + \overline{B} \quad (\text{Rule-1})$$

$$\overline{A + B} = \overline{A} \cdot \overline{B} \quad (\text{Rule-2})$$

- **How to remember?**
 - **break the line, change the sign**
- **Break the LINE over the two variables, and change the SIGN under the line.**

Complementing Functions (continued)

- Use **DeMorgan's Theorem** repeatedly to complement a function.

- Example: $F = \bar{x}y\bar{z} + x\bar{y}\bar{z}$

$$\begin{aligned}\bar{F} &= \overline{\bar{x}y\bar{z} + x\bar{y}\bar{z}} = \overline{\bar{x}y\bar{z}} \cdot \overline{x\bar{y}\bar{z}} \\ &= (x + \bar{y} + z)(\bar{x} + y + z)\end{aligned}$$

- Example: $G = (\bar{a} + bc)\bar{d} + e$

$$\begin{aligned}\bar{G} &= \overline{(\bar{a} + bc)\bar{d} + e} = \overline{((\bar{a} + bc)\bar{d})} \cdot \bar{e} \\ &= \overline{((\bar{a} + bc) + d)} \cdot \bar{e} = (a \cdot \bar{bc} + d) \cdot \bar{e} \\ &= (a \cdot (\bar{b} + \bar{c}) + d) \cdot \bar{e}\end{aligned}$$

Applications of Boolean Algebra

- **Boolean Function Evaluation**
- **Boolean Algebraic Proof**
- **Expression Simplification**

Boolean Function Evaluation

$$F1 = xy\bar{z}$$

$$F2 = x + \bar{y}z$$

$$F3 = \bar{x}\bar{y}\bar{z} + \bar{x}yz + x\bar{y}$$

$$F4 = x\bar{y} + \bar{x}z$$

x	y	z	F1	F2	F3	F4
0	0	0	0	0		
0	0	1	0	1		
0	1	0	0	0		
0	1	1	0	0		
1	0	0	0	1		
1	0	1	0	1		
1	1	0	1	1		
1	1	1	0	1		

Boolean Algebraic Proof: Example 1

- $A + A \cdot B = A$ (Absorption Theorem)

Proof Steps	Justification (identity or theorem)
-------------	-------------------------------------

$A + A \cdot B$	
-----------------	--

$= A \cdot 1 + A \cdot B$	$X = X \cdot 1$
---------------------------	-----------------

$= A \cdot (1 + B)$	$X \cdot Y + X \cdot Z = X \cdot (Y + Z)$ (Distributive Law)
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$= A \cdot 1$	$1 + X = 1$
---------------	-------------

$= A$	$X \cdot 1 = X$
-------	-----------------

- Our primary reason for doing proofs is to learn:
 - Careful and efficient use of the identities and theorems of Boolean algebra, and
 - How to choose the appropriate identity or theorem to apply to make forward progress, irrespective of the application.

Boolean Algebraic Proof: Example 2

- $AB + \bar{A}C + BC = AB + \bar{A}C$ (**Consensus Theorem**)

Proof Steps	Justification (identity or theorem)
-------------	-------------------------------------

$AB + \bar{A}C + BC$	
----------------------	--

$= AB + \bar{A}C + 1 \cdot BC$	
--------------------------------	--

$= AB + \bar{A}C + (A + \bar{A}) \cdot BC$	
--	--

$= AB + \bar{A}C + ABC + \bar{A}BC$	
-------------------------------------	--

$= (AB + ABC) + (\bar{A}C + \bar{A}BC)$	
---	--

$= AB(1+C) + \bar{A}C(1+B)$	
-----------------------------	--

$= AB + \bar{A}C$	
-------------------	--

Boolean Algebraic Proof: Example 3

- $(\overline{X + Y})Z + X\overline{Y} = \overline{Y}(X + Z)$

Proof Steps	Justification (identity or theorem)
-------------	-------------------------------------

$(\overline{X + Y})Z + X\overline{Y}$	
$= \overline{X}\overline{Y}Z + X\overline{Y}$	
$= \overline{Y}(\overline{X}Z + X)$	
$= \overline{Y}(\overline{X} + X)(Z + X)$	
$= \overline{Y} \cdot 1 \cdot (Z + X)$	
$= \overline{Y}(X + Z)$	

Proof of DeMorgan's Laws

$$\overline{X + Y} = \bar{X} \cdot \bar{Y}$$

$$\overline{X \cdot Y} = \bar{X} + \bar{Y}$$

- Using the Complementary and Substitution Theorem to prove De Morgan's Laws.
- To show this we need to show that $A + \bar{A} = 1$ and $A \cdot \bar{A} = 0$ with $A = X + Y$ and $\bar{A} = \bar{X} \cdot \bar{Y}$. This proves that $\overline{X + Y} = \bar{X} \cdot \bar{Y}$.
- Part 1: Show $X + Y + \bar{X} \cdot \bar{Y} = 1$
- Part 2: Show $(X + Y) \cdot (\bar{X} \cdot \bar{Y}) = 0$

Useful Theorems

- $x \cdot y + \bar{x} \cdot y = y$ $(x + y)(\bar{x} + y) = y$ **Minimization**
- $x + x \cdot y = x$ $x \cdot (x + y) = x$ **Absorption**
- $x + \bar{x} \cdot y = x + y$ $x \cdot (\bar{x} + y) = x \cdot y$ **Simplification**
- $x \cdot y + \bar{x} \cdot z + y \cdot z = x \cdot y + \bar{x} \cdot z$ **Consensus**
 $(x + y)(\bar{x} + z)(y + z) = (x + y)(\bar{x} + z)$
- $\overline{x + y} = \bar{x} \cdot \bar{y}$ $\overline{x \cdot y} = \bar{x} + \bar{y}$ **DeMorgan's Laws**

Expression Simplification

- An application of Boolean algebra
- Simplify to contain the smallest number of literals (complemented and uncomplemented variables):

$$\begin{aligned} & AB + \bar{A}CD + \bar{A}BD + \bar{A}C\bar{D} + ABCD \\ = & AB + ABCD + \bar{A}CD + \bar{A}C\bar{D} + \bar{A}BD \\ = & AB + AB(CD) + \bar{A}C(D + \bar{D}) + \bar{A}BD \\ = & AB + \bar{A}C + \bar{A}BD = B(A + \bar{A}D) + \bar{A}C \\ = & B(A + D) + \bar{A}C \qquad \qquad \qquad 5 \text{ literals} \end{aligned}$$

Example: Simplify Expression

$$L = AB + \overline{A}\overline{C} + \overline{B}C + \overline{C}B + \overline{B}D + \overline{D}B + ADE(F + G) \text{ —}$$

$$L = \overline{\overline{A}\overline{B}\overline{C}} + \overline{B}C + \overline{C}B + \overline{B}D + \overline{D}B + ADE(F + G) \quad \text{DeMorgan Laws}$$

$$= A + \overline{B}C + \overline{C}B + \overline{B}D + \overline{D}B + ADE(F + G) \quad A + \overline{A}B = A + B$$

$$= A + \overline{B}C + \overline{C}B + \overline{B}D + \overline{D}B \quad A + AB = A$$

$$= A + \overline{B}C(D + \overline{D}) + \overline{C}B + \overline{B}D + \overline{D}B(C + \overline{C}) \quad A + \overline{A} = 1$$

$$= A + \overline{B}CD + \overline{B}C\overline{D} + \overline{C}B + \overline{B}D + \overline{D}BC + \overline{D}B\overline{C} \quad \text{Distributive Laws}$$

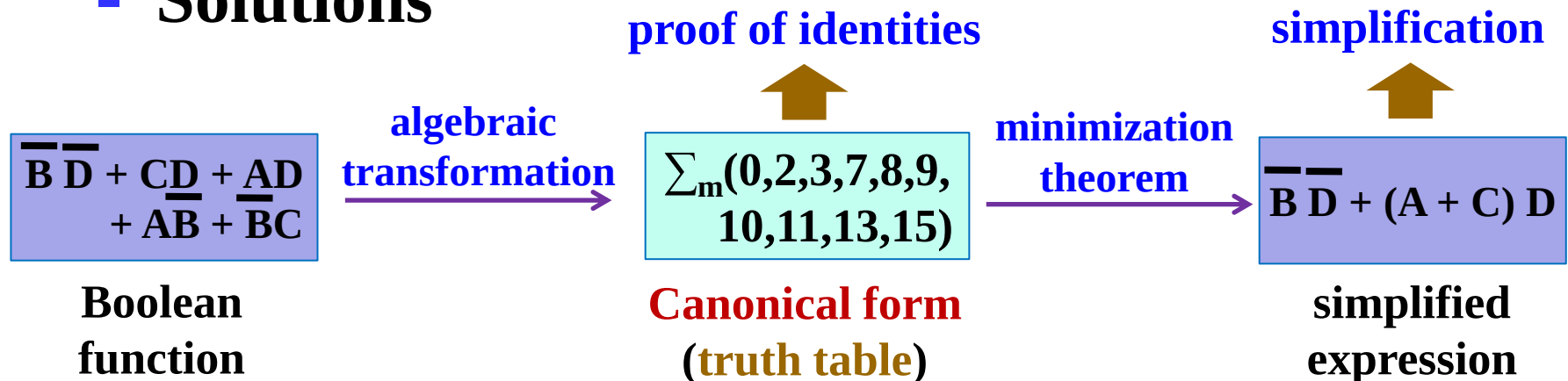
$$= A + \overline{B}C\overline{D} + \overline{C}B + \overline{B}D + \overline{D}BC \quad A + AB = A$$

$$= A + C\overline{D}(\overline{B} + B) + \overline{C}B + \overline{B}D$$

$$= A + C\overline{D} + \overline{C}B + \overline{B}D \quad A + \overline{A} = 1$$

Difficulties in Manipulating Boolean Functions

- For Boolean function proof
 - **Problem:** Boolean functions are not unique (e.g., $A B + A C + B C = (A+B)(A+C)(B+C)$).
- For expression simplification
 - **Problem 1:** too many identities and theorems to apply
 - **Problem 2:** can be tricky to simplify expressions (e.g., $BC = (A + \overline{A}) \cdot BC$)
- Solutions



Overview – Canonical Forms

- **What are Canonical Forms?**
- **Minterms and Maxterms**
- **Index Representation of Minterms and Maxterms**
- **Sum-of-Minterm (SOM) Representations**
- **Product-of-Maxterm (POM) Representations**
- **Representation of Complements of Functions**
- **Conversions between Representations**

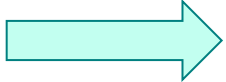
Canonical Forms

- It is useful to specify Boolean functions in a form that:
 - has a correspondence to the truth tables
 - allows comparison for equality
 - provides a starting point for optimization
- Two common Canonical Forms:
 - Sum of Minterms (SOM)
 - Product of Maxterms (POM)

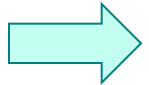
Boolean Expressions for a Truth Table (1/3)

X	Y	F	$\overline{F}$
0	0	1	0
0	1	0	1
1	0	1	0
1	1	0	1

- Three ways to express the truth table for F:

1. $F = \overline{X}\overline{Y} + X\overline{Y}$  Sum of Minterms (SOM)

2. $F = \overline{\overline{X}Y + X\overline{Y}} = (X + \overline{Y})(\overline{X} + \overline{Y})$

3. $F = (X + \overline{Y})(\overline{X} + \overline{Y})$  Product of Maxterms (POM)

Boolean Expressions for a Truth Table (2/3)

X	Y	F	$\overline{F}$
0	0	1	0
0	1	0	1
1	0	1	0
1	1	0	1

→ minterm: $\overline{X}\overline{Y}$

→ minterm: $X\overline{Y}$

■ Three ways to express the truth table for F:

1. $F = \overline{X}\overline{Y} + X\overline{Y} = \sum_m(0,2)$ → Sum of Minterms (SOM)
2. $F = \overline{X}Y + X\overline{Y} = (X + \overline{Y})(\overline{X} + \overline{Y})$
3. $F = (X + \overline{Y})(\overline{X} + \overline{Y}) = \prod_M(1,3)$ → Product of Maxterms (POM)

Boolean Expressions for a Truth Table (3/3)

	X	Y	F	$\overline{F}$
maxterm $X + \overline{Y}$	0	0	1	0
	0	1	0	1
	1	0	1	0
	1	1	0	1

maxterm $\overline{X} + \overline{Y}$

■ Three ways to express the truth table F:

1. $F = \overline{X}\overline{Y} + X\overline{Y} = \sum_m(0,2)$ $\Rightarrow$ Sum of Minterms (SOM)
2. $F = \overline{X}Y + X\overline{Y} = (X + \overline{Y})(\overline{X} + \overline{Y})$
3. $F = (X + \overline{Y})(\overline{X} + \overline{Y}) = \prod_M(1,3)$ $\Rightarrow$ Product of Maxterms (POM)

Minterms

- **Minterms** are **AND terms** with every variable present in either **true** (e.g., x) or **complemented form** (e.g., $\overline{x}$).
- For n -input variables, there are 2^n minterms or 2^n total number of possible input combinations.
- Example: Two variables (X and Y) produce $2 \times 2 = 4$ combinations:
 XY (both normal)
 $X\overline{Y}$ (X normal, Y complemented)
 $\overline{X}Y$ (X complemented, Y normal)
 $\overline{X}\overline{Y}$ (both complemented)
- Thus there are four minterms of two variables.

Minterms

- Each minterm equals 1 at exactly one particular input combination and is equal 0 at all other combinations.
- For example, $\bar{x}\bar{y}\bar{z}$ is always equal to 0 except for the input combination $xyz=000$, where it is equal to 1.
- Accordingly, the minterm $\bar{x}\bar{y}\bar{z}$ is referred to as m_0 .
- In general, minterms are designated m_i , where i corresponds the input combination at which this minterm is equal to 1.

Maxterms

- **Maxterms** are **OR terms** with every variable present in either **true** (e.g., x) or **complemented form** (e.g., $\bar{x}$).
- For n -input variables, there are 2^n total number of possible input combinations or 2^n maxterms.
- Example: Two variables (X and Y) produce $2 \times 2 = 4$ combinations:

$X + Y$ (both normal)

$X + \bar{Y}$ (x normal, y complemented)

$\bar{X} + Y$ (x complemented, y normal)

$\bar{X} + \bar{Y}$ (both complemented)

Maxterms

- Each maxterm equals 0 at exactly one particular input combination and is equal 1 at all other combinations.
- For example, $x+y+z$ is always equal to 1 except for the input combination $xyz=000$, where it is equal to 0.
- Accordingly, the maxterm $x+y+z$ is referred to as M_0 .
- In general, maxterms are designated M_i , where i corresponds the input combination at which this maxterm is equal to 0.

Standard Order

- Minterms and maxterms are designated with a subscript
- The subscript is a number, corresponding to a binary pattern
- The bits in the pattern represent the complemented or normal state of each variable listed in a standard order.
- All variables will be present in a minterm or maxterm and will be listed in the same order (usually **alphabetically**)
- Example: For variables a, b, c:
 - Maxterms: $(a + b + \bar{c})$, $(a + b + c)$
 - Terms: $(b + a + c)$, $a \bar{c} b$, and $(c + b + a)$ are NOT in standard order.
 - Minterms: $a \bar{b} c$, $a b c$, $\bar{a} \bar{b} c$
 - Terms: $(a + c)$, $\bar{b} c$, and $(\bar{a} + b)$ do not contain all variables

Index for Maxterms and Minterms

- **Examples: Two variable minterms and maxterms.**

x y	Index₁₀	Minterm	Maxterm
0 0	0	$\bar{x} \bar{y}$	$x + y$
0 1	1	$\bar{x} y$	$x + \bar{y}$
1 0	2	$x \bar{y}$	$\bar{x} + y$
1 1	3	$x y$	$\bar{x} + \bar{y}$

- **The index above is important for describing which variables in the terms are true and which are complemented.**

Purpose of the Index

- The **index** for the minterm or maxterm, expressed as a **binary number**, is used to determine whether the variable is shown in the true form or complemented form.
- For Minterms (base 2):
 - “0” means the variable is “Complemented” and
 - “1” means the variable is “Not Complemented”.
- For Maxterms (base 2):
 - “0” means the variable is “Not Complemented” and
 - “1” means the variable is “Complemented”.

Index Example in Three Variables

- **Example:** (for three variables)
- Assume the variables are called X, Y, and Z.
- The standard order is X, then Y, then Z.
- The Index 0 (base 10) = 000 (base 2) for three variables). All three variables are complemented for minterm 0 ($\bar{X} \cdot \bar{Y} \cdot \bar{Z}$) and no variables are complemented for Maxterm 0 ($X+Y+Z$).
 - Minterm 0, called m_0 is $\bar{X}\bar{Y}\bar{Z}$.
 - Maxterm 0, called M_0 is $(X + Y + Z)$.
 - Minterm 6 ? $XY\bar{Z}$
 - Maxterm 6 ? $(\bar{X} + \bar{Y} + Z)$

Index Examples – Four Variables

Index	Binary	Minterm	Maxterm
i	Pattern	m_i	M_i
0	0000	$\bar{a}\bar{b}\bar{c}\bar{d}$	$a + b + c + d$
1	0001	$\bar{a}\bar{b}\bar{c}d$	?
3	0011	?	$a + b + \bar{c} + \bar{d}$
5	0101	$\bar{a}b\bar{c}d$	$a + \bar{b} + c + \bar{d}$
7	0111	?	$a + \bar{b} + \bar{c} + \bar{d}$
10	1010	$a\bar{b}c\bar{d}$	$\bar{a} + b + \bar{c} + d$
13	1101	$ab\bar{c}d$	?
15	1111	$abcd$	$\bar{a} + \bar{b} + \bar{c} + \bar{d}$

Minterm and Maxterm Relationship

- Two-variable example:

$$M_2 = \bar{x} + y \text{ and } m_2 = x \bar{y}$$

- Review: DeMorgan's Theorem

$$\overline{x \cdot y} = \bar{x} + \bar{y} \text{ and } \overline{x + y} = \bar{x} \cdot \bar{y}$$

Thus M_2 is the complement of m_2 and vice-versa.

- Since DeMorgan's Theorem holds for n variables, the above holds for terms of n variables
- giving: $M_i = \overline{m_i}$ and $m_i = \overline{M_i}$

Inputs		Outputs	
X	Y	$X\bar{Y}$	$\bar{X}+Y$
0	0	0	1
0	1	0	1
1	0	1	0
1	1	0	1

Thus M_i is the complement of m_i .

Function Tables for Both

- **Minterms of 2 variables**

x y	m_0	m_1	m_2	m_3
0 0	1	0	0	0
0 1	0	1	0	0
1 0	0	0	1	0
1 1	0	0	0	1

Maxterms of 2 variables

x y	M_0	M_1	M_2	M_3
0 0	0	1	1	1
0 1	1	0	1	1
1 0	1	1	0	1
1 1	1	1	1	0

- Each column in the maxterm function table is the complement of the column in the minterm function table since M_i is the complement of m_i .

Observations

- In the function tables:
 - Each minterm has **one and only one 1** present in the 2^n terms (a minimum of 1s). All other entries are 0.
 - Each maxterm has **one and only one 0** present in the 2^n terms. All other entries are 1 (a maximum of 1s).
- We can implement any function by "ORing" the **minterms corresponding to "1" entries in the function table**. These are called the **minterms of the function**.
- We can implement any function by "ANDing" the **maxterms corresponding to "0" entries in the function table**. These are called the **maxterms of the function**.
- This gives us two canonical forms:
 - Sum of Minterms (SOM)
 - Product of Maxterms (POM)for stating any Boolean function.

Minterm Function Example

- Example: Find $F(A,B,C) = m_1 + m_4 + m_7$
- $F(A,B,C) = \bar{A} \bar{B} C + A \bar{B} \bar{C} + ABC$



A B C	index	m_1	+	m_4	+	m_7	= F_1
0 0 0	0	0	+	0	+	0	= 0
0 0 1	1	1	+	0	+	0	= 1
0 1 0	2	0	+	0	+	0	= 0
0 1 1	3	0	+	0	+	0	= 0
1 0 0	4	0	+	1	+	0	= 1
1 0 1	5	0	+	0	+	0	= 0
1 1 0	6	0	+	0	+	0	= 0
1 1 1	7	0	+	0	+	1	= 1

Minterm Function Example

- $F(A, B, C, D, E) = m_2 + m_9 + m_{17} + m_{23}$
- $F(A, B, C, D, E) = \overline{A}\overline{B}\overline{C}D\overline{E} + \overline{A}B\overline{C}\overline{D}E + \overline{A}\overline{B}\overline{C}\overline{D}E + A\overline{B}CDE$

Maxterm Function Example

- Example: Implement $F_1(x,y,z)$ in maxterms:

$$F_1 = M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6$$

$$F_1 = (x + y + z) (x + \bar{y} + z) (x + \bar{y} + \bar{z}) \\ (\bar{x} + y + \bar{z}) (\bar{x} + \bar{y} + z)$$

x y z	i	$M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6 = F_1$
0 0 0	0	0 · 1 · 1 · 1 · 1 = 0
0 0 1	1	1 · 1 · 1 · 1 · 1 = 1
0 1 0	2	1 · 0 · 1 · 1 · 1 = 0
0 1 1	3	1 · 1 · 0 · 1 · 1 = 0
1 0 0	4	1 · 1 · 1 · 1 · 1 = 1
1 0 1	5	1 · 1 · 1 · 0 · 1 = 0
1 1 0	6	1 · 1 · 1 · 1 · 0 = 0
1 1 1	7	1 · 1 · 1 · 1 · 1 = 1



SOM and POM in Our Daily Life

- Suppose the class need to schedule a meeting and you have to report your class schedule for Friday.
 - Set logic variable **isAvailable** (1: available / 0: not available)

	Monday	Tuesday	Wednesday	Thursday	Friday
7 ^{AM} :30					
8 ^{AM} :30	MATH	SCIENCE	MATH	ENGLISH	BAND
9 ^{AM} :30				READING	
10 ^{AM} :30	SOCIAL STUDIES				
11 ^{AM} :30		ENGLISH	SPANISH	HISTORY	SOCIAL STUDIES
12 ^{PM} :30	LUNCH	LUNCH	LUNCH	LUNCH	LUNCH
1 ^{PM} :30	PE	HISTORY	PE	SCIENCE	PE
2 ^{PM} :30	ENGLISH	BAND	SOCIAL STUDIES	BAND	MATH
3 ^{PM} :30					
4 ^{PM} :30					
5 ^{PM} :30					
6 ^{PM} :30					
7 ^{PM} :30					

→ Maxterm
→ Minterm

- report solution 1: 7:00-7:30, 9:00-10:00, 16:00-20:00 **available**
- report solution 2: 7:30-9:00, 10:00-16:00 **not available**

Transformation to Canonical Sum of Minterms

- **Any Boolean function can be expressed as a Sum of Minterms.**
 - For the function table, the minterms used are the terms corresponding to the 1's
 - For expressions, **expand all terms** first to explicitly list all minterms. Do this by “ANDing” any term **missing a variable v with a term $(v + \bar{v})$** .
- **Example: Implement $f = x + \bar{x} \bar{y}$ as a sum of minterms with two variables.**

First expand terms: $f = x(y + \bar{y}) + \bar{x} \bar{y}$

Then distribute terms: $f = xy + x\bar{y} + \bar{x} \bar{y}$

Express as sum of minterms: $f = m_0 + m_2 + m_3$

Another SOM Example

- **Example:** $F = A + \bar{B} C$
- **There are three variables, A, B, and C which we take to be the standard order.**
- **Expanding the terms with missing variables:**
$$\begin{aligned} F &= A(B + \bar{B})(C + \bar{C}) + (A + \bar{A}) \bar{B} C \\ &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} \\ &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C \\ &= m_7 + m_6 + m_5 + m_4 + m_1 = m_1 + m_4 + m_5 + m_6 + m_7 \end{aligned}$$
- **Collect terms (removing all but one of duplicate terms):**
- **Express as SOM:** $F = m_1 + m_4 + m_5 + m_6 + m_7$

Shorthand SOM Form

- From the previous example, we started with:

$$F = A + \bar{B} C$$

- We ended up with:

$$F = m_1 + m_4 + m_5 + m_6 + m_7$$

- This can be denoted in the formal shorthand:

$$F(A, B, C) = \Sigma_m(1, 4, 5, 6, 7)$$

- Note that we explicitly show the standard variables in order and drop the “m” designators.

Transformation to Canonical Product of Maxterms

- Any Boolean Function can be expressed as a Product of Maxterms (POM).
 - For the function table, the maxterms used are the terms corresponding to the 0's.
 - For an expression, expand all terms first to explicitly list all maxterms. Do this by first applying the second **distributive law**, “ORing” terms **missing variable** v with a term equal to and $v \cdot \bar{v}$ then applying the **distributive law** again.

- Example: Convert to product of maxterms:

$$f(x, y, z) = x + \bar{x} \bar{y}$$

Apply the distributive law:

$$x + \bar{x} \bar{y} = (x + \bar{x})(x + \bar{y}) = 1 \cdot (x + \bar{y}) = x + \bar{y}$$

Add missing variable z :

$$x + \bar{y} + z \cdot \bar{z} = (x + \bar{y} + z)(x + \bar{y} + \bar{z})$$

Express as POM: $f = M_2 \cdot M_3$

Another POM Example

- Convert to Product of Maxterms:

$$f(A, B, C) = A \bar{C} + BC + \bar{A} \bar{B}$$

- Use $x + yz = (x+y)(x+z)$ with $x = (A \bar{C} + BC)$, $y = \bar{A}$, and $z = \bar{B}$ to get:

$$f = (A \bar{C} + BC + \bar{A})(A \bar{C} + BC + \bar{B})$$

- Then use $x + \bar{x}y = x + y$ to get:

$$f = (\bar{C} + BC + \bar{A})(A \bar{C} + C + \bar{B})$$

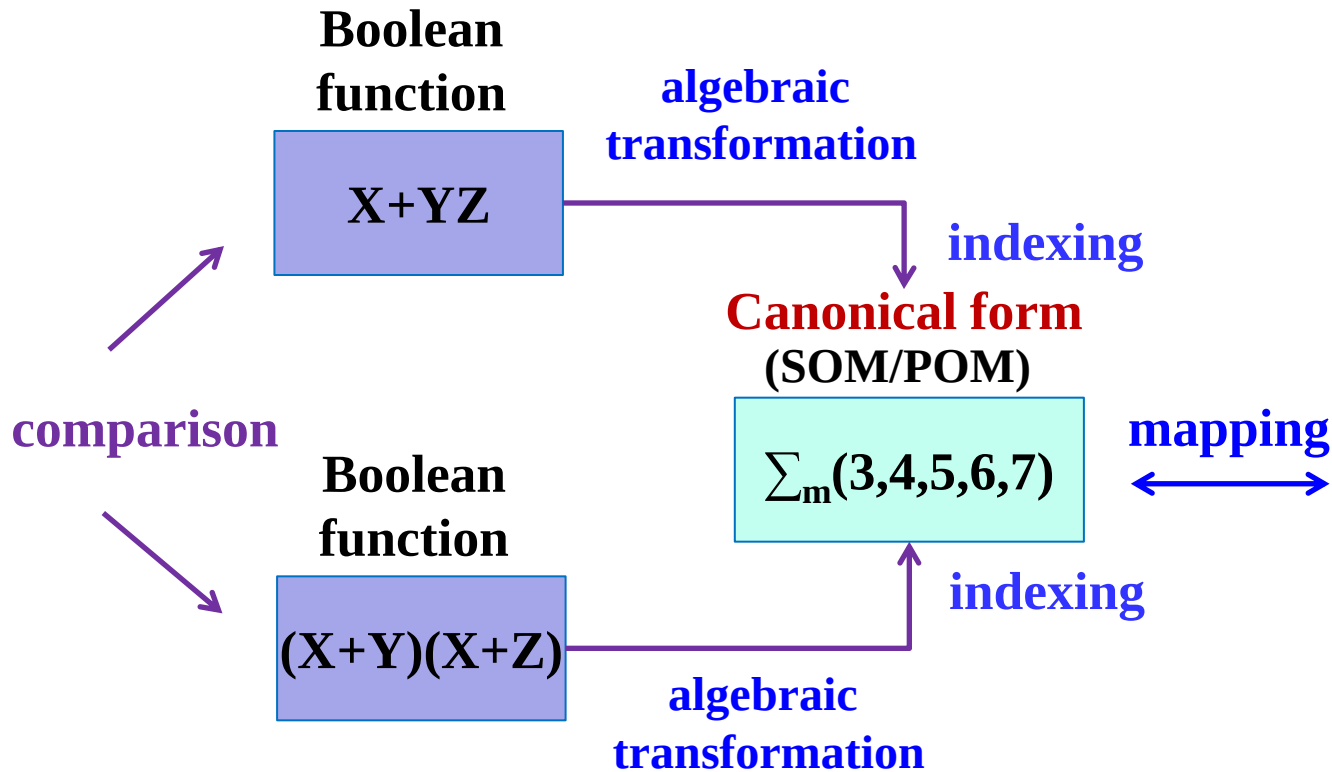
and a second time to get:

$$f = (\bar{C} + B + \bar{A})(A + C + \bar{B})$$

- Rearrange to standard order,

$$f = (\bar{A} + B + \bar{C})(A + \bar{B} + C) \text{ to give } f = M_2 \cdot M_5$$

Canonical Forms for Comparison of Equality



Truth table

X	Y	Z	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Function Complements

- The **complement of a function** expressed as a sum of minterms is constructed by selecting the minterms **missing** in the **sum-of-minterms** canonical forms.
- Alternatively, the complement of a function expressed by a Sum of Minterms form is simply the **Product of Maxterms** with the **same** indices.
- Example: Given $F(x, y, z) = \Sigma_m(1, 3, 5, 7)$

$$\bar{F}(x, y, z) = \Sigma_m(0, 2, 4, 6)$$

$$\bar{F}(x, y, z) = \Pi_M(1, 3, 5, 7)$$

X	Y	Z	F	$\bar{F}$
0	0	0	0	1
0	0	1	1	0
...	...	...	...	...

Conversion Between Forms

- To convert between sum-of-minterms and product-of-maxterms form (or vice-versa) :
 - Identify the terms that are NOT included in the expression.
 - Combine all the terms using the corresponding form.
- Example: $F(x, y, z) = \Sigma_m(1,3,5,7)$, the product of maxterms form can be obtained as follows:
 - Identify the missing minterms: 0, 2, 4 and 6.
 - Use product-of-maxterms to combine all the maxterms.
$$F(x, y, z) = \Pi_M(0,2,4,6)$$
- Converting maxterms to minterms has the same process.

Minterms to Maxterms Conversion

X	Y	Z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$$F(X,Y,Z) = \Sigma_m(1,3,5,7)$$

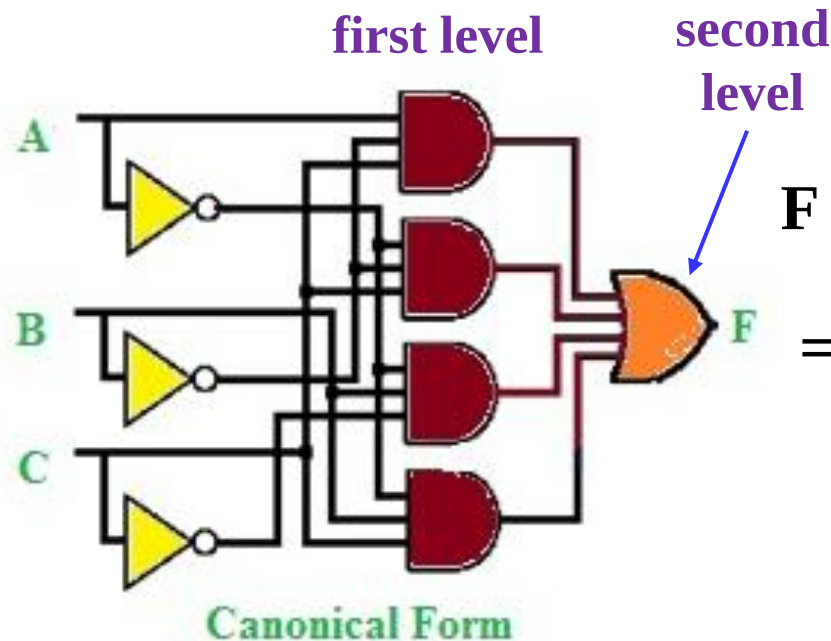
- **Example: Implement F in maxterms:**

$$F = \Sigma_m(1,3,5,7) = m_1 + m_3 + m_5 + m_7$$

$$F = \Pi_M(0,2,4,6) = M_0 \cdot M_2 \cdot M_4 \cdot M_6$$

Two-level Logic with Canonical Form

- A sum of minterms form for n variables can be written down directly from a truth table.
 - Implementation of this form is a **two-level AND-OR network** such that:
 - The first level consists of n -input **AND gates**
 - The second level is a **single OR gate**



$$F(A, B, C) = \Sigma_m(1, 2, 3, 5)$$
$$= \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}C$$

Standard Forms

- The two canonical forms are basic forms that can be obtained from reading a given function's truth table.
- Canonical forms are very seldom the ones with the least number of literals, because each minterm or maxterm must contain all the variables.
- Another way to express Boolean functions is in standard form.
- In standard forms, **the terms may contain one, two or any number of literals.**

Standard Forms (continued)

- Standard Sum-of-Products (SOP) form:
equations are written as an OR of AND terms
- Standard Product-of-Sums (POS) form:
equations are written as an AND of OR terms
- Examples:
 - SOP: $A B C + \bar{A} \bar{B} C + B$
 - POS: $(A + B) \cdot (A + \bar{B} + \bar{C}) \cdot C$
- These “mixed” forms are neither SOP nor POS
 - $(A \bar{B} + C) (A + C)$
 - $A \bar{B} C + A C (A + B)$

Standard Sum-of-Products (SOP)

- A Simplification Example:

- $F(A, B, C) = \sum m(1, 4, 5, 6, 7)$

- Writing the minterm expression:

$$F = \overline{A} \overline{B} C + A \overline{B} \overline{C} + A \overline{B} C + A B \overline{C} + A B C$$

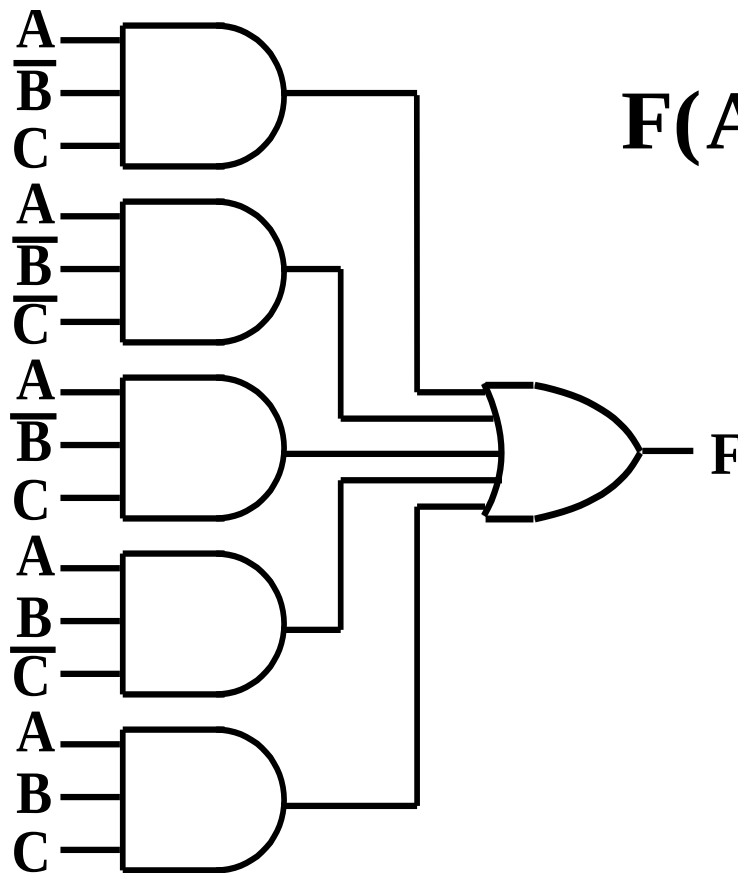
- Simplifying:

$$\begin{aligned} F &= \overline{A} \overline{B} C + (\overline{A} \overline{B} \overline{C} + \overline{A} B \overline{C}) + (A B \overline{C} + A B C) \\ &= \overline{A} \overline{B} C + (\overline{A} \overline{B} + A B) \\ &= \overline{A} \overline{B} C + A \\ &= \overline{B} C + A \end{aligned}$$

- Simplified F contains 3 literals compared to 15 in minterm F

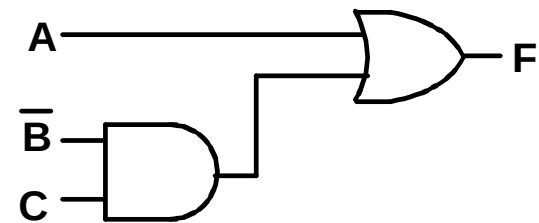
AND/OR Two-level Implementation of SOP Expression

- The two implementations for F are shown below – it is quite apparent which is simpler!



Canonical Form

$$F(A, B, C) = \Sigma m(1, 4, 5, 6, 7)$$



$$F = \overline{B}C + A$$

Standard SOP Form

SOP and POS Observations

- **The previous examples show that:**
 - **Canonical Forms (Sum-of-minterms, Product-of-Maxterms), or other standard forms (SOP, POS) differ in complexity**
 - **Boolean algebra can be used to manipulate equations into simpler forms.**
 - **Simpler equations lead to simpler two-level implementations**
- **Questions:**
 - **How to attain a “simplest” expression?**
 - **How to measure the simplicity of a circuit?**
 - **Is there only one minimum cost circuit?**
 - **The next part will deal with these issues.**

Assignment

Reading:

■ 2.1-2.3

Problem assignment:

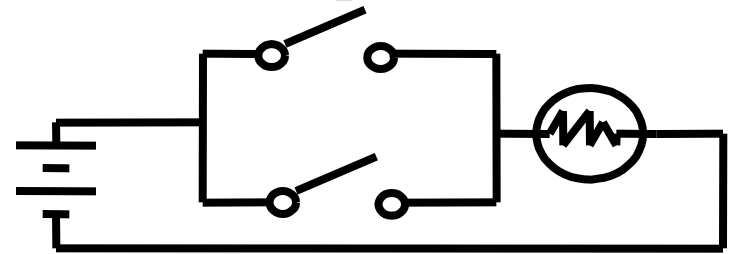
■ 2-1a; 2-2a, c; 2-3a, c; 2-6b, d; 2-10a, c; 2-11a, c, e; 2-12b

Logic Function Implementation

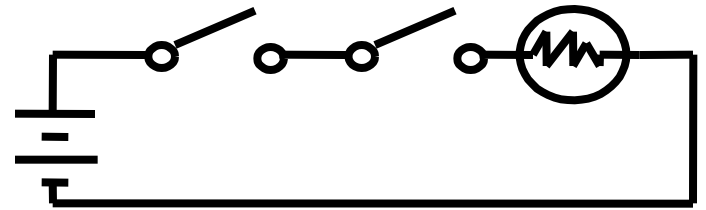
■ Using Switches

- For **inputs**:
 - logic 1 is switch closed
 - logic 0 is switch open
- For **outputs**:
 - logic 1 is light on
 - logic 0 is light off.
- NOT uses a switch such that:
 - logic 1 is switch open
 - logic 0 is switch closed

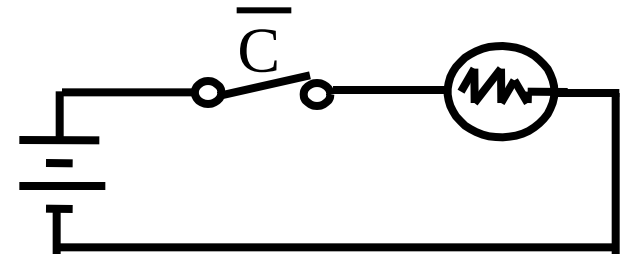
Switches in parallel => OR



Switches in series => AND

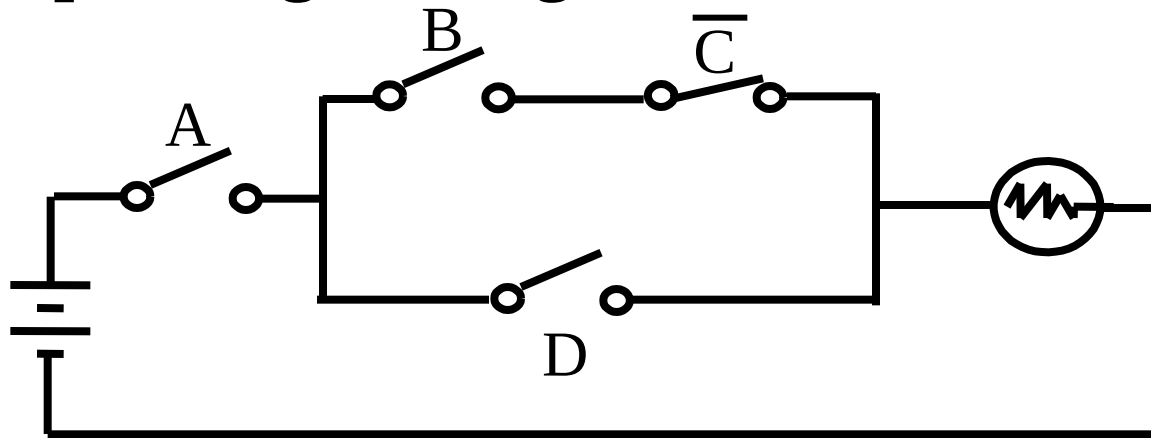


Normally-closed switch => NOT



Logic Function Implementation (Continued)

- **Example: Logic Using Switches**



- **Light is on ($L = 1$) for**

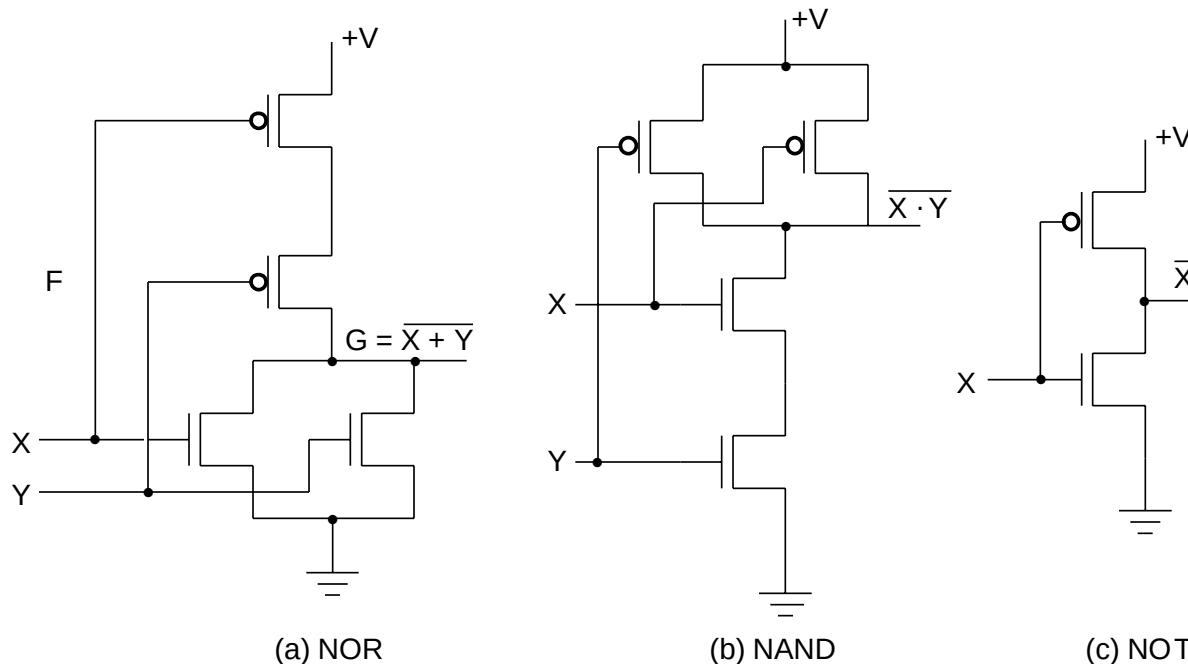
$$L(A, B, C, D) = A ((B \overline{C}) + D) = A B \overline{C} + A D$$

and off ($L = 0$), otherwise.

- **Useful model for relay circuits and for CMOS gate circuits, the foundation of current digital logic technology**

Appendix: Logic Gates (Continued)

- Implementation of logic gates with CMOS Circuits



- Transistor or tube implementations of logic functions are called logic gates or just gates
- Transistor gate circuits can be modeled by switch circuits

Conversion Between Forms

- To convert between sum-of-minterms and product-of-maxterms form (or vice-versa) we follow these steps:
 - Find the function **complement** by swapping terms in the list with terms not in the list.
 - **Change** from products to sums, or vice versa.
- Example: Given F as : $F(x, y, z) = \sum_m(1, 3, 5, 7)$
- Form the Complement: $\bar{F}(x, y, z) = \sum_m(0, 2, 4, 6)$
- Then use the other form with the same indices – this forms the complement again, giving the other form of the original function: $F(x, y, z) = \prod_M(0, 2, 4, 6)$

Minterms to Maxterms Conversion

X	Y	Z	F	$\bar{F}$
0	0	0	0	1
0	0	1	1	0
0	1	0	0	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	0

$$F(X,Y,Z) = \Sigma_m(1,4,7)$$

- Example: Implement F1 in maxterms:

$$\bar{F} = m_0 + m_2 + m_3 + m_5 + m_6$$

$$F = \bar{m}_0 \cdot \bar{m}_2 \cdot \bar{m}_3 \cdot \bar{m}_5 \cdot \bar{m}_6$$

$$= M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6$$

DeMorgan Laws